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COOKING VESSEL

Abstract

A cooking vessel has a cooking body and a lid. The cooking body defines a cooking chamber. The lid is removably couplable to the cooking body. The lid has a lid body and a liquid reservoir. The liquid reservoir is provided within the lid body. The at least one sensor is configured to measure at least one of a humidity or a temperature of the cooking chamber.

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Background/Summary

TECHNICAL FIELD

[0001] The description generally relates to a cooking vessel, and more specifically to a cooking vessel suitable for use with a cooking appliance.

BACKGROUND

[0002] Cooking appliances, such as cooktops, ranges, etc., have cook surfaces with hobs that generate heat for cooking or warming food items in a cooking vessel having a cooking chamber. Traditional cooking appliances provide thermal energy to the hob, such as by a gas fuel or electricity, which is conductively transferred to a cooking vessel placed thereon. In other examples, induction cooking appliances provide energy directly to the cooking vessel by way of electromagnetic induction using a variable-current coil and a suitable metallic cooking vessel. During operation, such induction cooking appliances can have much cooler surface temperatures compared to traditional cooking appliances with conductive hobs or burners.

BRIEF SUMMARY

[0003] In one aspect, the disclosure relates to a cooking vessel comprising a cooking body defining a cooking chamber, a lid removably couplable to the cooking body, the lid having a lid body extending between an inner surface confronting the cooking chamber, and an outer surface, the lid body having a centerline axis extending between the inner surface and the outer surface, and a liquid reservoir provided within the lid body, and at least one sensor configured to measure at least one of a humidity or a temperature of the cooking chamber wherein a liquid within the liquid reservoir is selectively supplied to the cooking chamber based on at least one of the measured humidity of temperature of the cooking chamber.

[0004] In another aspect, the disclosure relates a cooking appliance comprising a cook surface with a heating zone configured to receive a cooking vessel, the cooking vessel comprising a cooking body defining a cooking chamber, a lid removably couplable to the cooking body, the lid having a lid body extending between an inner surface confronting the cooking chamber, and an outer surface, the lid body having a centerline axis extending between the inner surface and the outer surface, and at least one sensor configured to measure at least one of a humidity or a temperature of the cooking chamber, wherein a temperature of the heating zone is configured to be adjusted based on at least one of the humidity, the temperature, or a combination thereof of the cooking chamber.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] In the drawings:

[0006] FIG. 1 is a perspective schematic view of a cooking appliance and a cooking vessel having a cooking body and a lid, the cooking body defining a cooking chamber, in accordance with various aspects described herein.

[0007] FIG. 2 is a schematic cross-sectional view of the cooking appliance and the cooking vessel of FIG. 1, further illustrating at least one sensor provided within the lid, the lid having an inner surface confronting the cooking chamber, in accordance with various aspects described herein.

[0008] FIG. 3 is a schematic view of the inner surface of the lid as seen from sight-line III-III of FIG. 2, further illustrating a set of outlets provided along the inner surface, in accordance with various aspects described herein.

[0009] FIG. 4 is a schematic view of a lid of an exemplary cooking vessel suitable for use as the cooking vessel of FIG. 1, further illustrating a set of outlets provided along the inner surface, in accordance with various aspects described herein.

[0010] FIG. 5 is a schematic cross-sectional side view of a lid of an exemplary cooking vessel suitable for use as the cooking vessel of FIG. 1, further illustrating a distribution duct provided within the lid.

DETAILED DESCRIPTION

[0011] A cooking vessel includes a cooking body defining a cooking chamber. The cooking vessel includes a lid. The lid is removably couplable to the cooking body. The lid provides selective access to the cooking chamber. The cooking vessel includes a sensor that is configured to measure at least one of a humidity or a temperature of contents (e.g., food items) of the cooking chamber. The cooking vessel is suitable for use with a cooking appliance having at least one heating zone. The at least one heating zone includes a heating element that generates a heat and transfers thermal energy to the cooking chamber to heat the contents of the cooking chamber. As used herein, the cooking appliance is any suitable device or element that is configured to produce a heat that is at least partially transferable to the cooking chamber. At least one of a liquid can be dispensed into the cooking chamber, the temperature of the heating zone can be changed, or a combination thereof can be done based on a measurement from the sensor.

[0012] The cooking appliance, as described herein, can be an induction cooking appliance. It will be appreciated, however, that the cooking vessel is suitable for use within any suitable cooking appliance such as, but not limited to, an oven, a microwave, a stovetop range, an induction stovetop, a fire (e.g., a camp fire), a hot pad, a portable stove, a grill, a smoker, or the like.

[0013] Features, advantages, and aspects of the present disclosure are set forth or apparent from a consideration of the following detailed description, drawings, and claims. Moreover, the following detailed description is exemplary and intended to provide explanation without limiting the scope of the disclosure as claimed.

[0014] As used herein, the terms “first,” “second,” “third,” “fourth,” or the like can be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. In addition, the singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0015] Also, as used herein, while sensors can be described as “sensing” or “measuring” a respective value, sensing or measuring can include determining a value indicative of or related to the respective value, rather than directly sensing or measuring the value itself. The sensed or measured values can further be provided to additional components. For instance, the value can be provided to a controller module or processor, and the controller module or processor can perform processing on the value to determine a representative value or an electrical characteristic representative of said value.

[0016] Additionally, as used herein, a “controller” or “controller module” can include a component configured or adapted to provide instruction, control, operation, or any form of communication for operable components to effect the operation thereof. A controller module can include any known processor, microcontroller, or logic device, including, but not limited to: field programmable gate arrays (FPGA), an application specific integrated circuit (ASIC), a proportional controller (P), a proportional integral controller (PI), a proportional derivative controller (PD), a proportional integral derivative controller (PID controller), a hardware-accelerated logic controller (e.g. for encoding, decoding, transcoding, etc.), or the like, or a combination thereof. Non-limiting examples of a controller module can be configured or adapted to run, operate, or otherwise execute program code to effect operational or functional outcomes, including carrying out various methods, functionality, processing tasks, calculations, comparisons, sensing or measuring of values, or the like, to enable or achieve the technical operations or operations described herein. The operation or functional outcomes can be based on one or more inputs, stored data values, sensed or measured values, true or false indications, or the like. While “program code” is described, non-limiting examples of operable or executable instruction sets can include routines, programs, objects, components, data structures, algorithms, etc., that have the technical effect of performing particular tasks or implement particular abstract data types.

[0017] Such a controller module as described herein may also compare a first value with a second value and may operate or control operations of additional components based on the satisfaction of

that comparison. As used herein, the term “satisfies” or “satisfaction” of a comparison between a first value and a second value will refer to a determination of whether the first value exceeds the second value, or does not exceed the second value, or is equal to the second value, such that the comparison is “true” when satisfied. In addition, as used herein, the term “satisfies” or “satisfaction” of a comparison between a first value and a value range refers to a determination that the first value is within the value range, such that the comparison is “true” when satisfied. It will be understood that such a determination may easily be altered to be satisfied by a positive/negative comparison or a true/false comparison. Example comparisons can include comparing a sensed or measured value to a threshold value, or to a threshold value range. For example, when a sensed, measured, or provided value is compared with another value or range, including a stored or predetermined value or range, the satisfaction of that comparison can result in actions, functions, or operations controllable by the controller module.

[0018] Such a controller module as described herein can further include a data storage component accessible by the processor, including memory, whether transient, non-transient, volatile, or non-volatile memory. Machine-executable instructions, associated data structures, and program modules represent examples of program code for executing methods disclosed herein. Machine-executable instructions can include, for example, instructions and data, which cause a general purpose computer, special purpose computer, or special purpose processing machine to perform a certain function or group of functions.

[0019] Additional non-limiting examples of the memory can include Random Access Memory (RAM), Read-Only Memory (ROM), flash memory, or one or more different types of portable electronic memory, such as discs, DVDs, CD-ROMs, flash drives, universal serial bus (USB) drives, the like, or any suitable combination of these types of memory. In one example, the program code can be stored within the memory in a machine-readable format accessible by the processor. Additionally, the memory can store various data, data types, sensed or measured data values, inputs, generated or processed data, or the like, accessible by the processor in providing instruction, control, or operation to effect a functional or operable outcome, as described herein.

[0020] Here and throughout the specification and claims, range limitations are combined, and interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

[0021] All directional references (e.g., radial, axial, proximal, distal, upper, lower, upward, downward, left, right, lateral, front, back, top, bottom, above, below, vertical, horizontal, clockwise, counterclockwise, upstream, downstream, forward, aft, etc.) are only used for identification purposes to aid the reader's understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of aspects of the disclosure described herein. Connection references (e.g., attached, coupled, secured, fastened, connected, and joined) are to be construed broadly and can include intermediate members between a collection of elements and relative movement between elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and in fixed relation to one another. The exemplary drawings are for purposes of illustration only and the dimensions, positions, order and relative sizes reflected in the drawings attached hereto can vary.

[0022] “Hydraulic diameter” (D_h) as used herein is in reference to a hydraulic diameter of one or more cavities or openings (e.g., an outlet of a fuel nozzle) of a finished (e.g., manufactured) fuel nozzle. Hydraulic diameter is a commonly used term when handling flow in non-circular tubes and channels. When the cross-section is uniform along the tube or channel length, it is defined as $D_{sub.H} = 4a/p$ where “a” is the cross-sectional area of the flow and “p” is the wetted perimeter of the cross-section. The hydraulic diameter can further be indirectly related to the Reynolds number of the fluid flow. As such, the hydraulic diameter can be used to at least partially quantify the flow of fluid through an area or pipe. It will be appreciated that the specific calculations for the

hydraulic diameter are known and referenced herein without direct reference.

[0023] FIG. 1 is a schematic presentation of a cooking system **10** including a cooking appliance **12** and a cooking vessel **20**. As illustrated, the cooking appliance **12** is an induction cooking appliance. However, it will be appreciated that the cooking appliance **12** is any suitable cooking appliance. The cooking appliance **12** includes a cook surface **14** including at least one heating zone **16**. The heating zone **16** is any suitable area along, within or near the cooking appliance **12** where the cooking vessel **20** can be placed such that a heat from the cooking appliance **12** is transferrable to the cooking vessel **20**. While illustrated as being along the cooking appliance **12**, the heating zone **16** can be an interior portion of the cooking appliance **12** or an area near the cooking appliance **12**. As a non-limiting example, the cooking appliance **12** can be an oven having an interior cooking chamber that defines the heating zone **16**. While a single heating zone **16** is illustrated in FIG. 1, the cooking appliance **12** can include multiple heating zones **16** accommodating multiple cooking vessels **20**.

[0024] The cooking appliance **12** includes a heating element **50**. The heating element **50** is configured to generate heat within the cooking vessel **20** when the cooking vessel **20** is positioned over the heating zone **16**.

[0025] The cooking appliance **12** includes a user interface **18**. The user interface **18** is any suitable interface that is used by a user to control operation of the cooking appliance **12**. The control of the cooking appliance **12** can include, for example, setting a cooking mode, setting a cooking temperature, or setting a timer, or the like. As a non-limiting example, the user interface **18** can include any suitable interface elements such as buttons, knobs, toggles, display screens, touch screens, light sources, or audio sources, in some examples. As shown, the user interface **18** is located on the cooking appliance **12** though this need not be the case. In some implementations, the user interface **18** can be located on a separate device, such as a mobile device, a tablet, or a central hub in some non-limiting examples. It is also contemplated that the user interface **18** can include user-interactive elements on the cooking appliance **12** and additional user-interactive elements on a separate device.

[0026] The cooking vessel **20** includes a cooking body **22** and a lid **24**. The cooking body **22** defines a cooking chamber **26**. The lid **24** includes an inner surface **28** and an outer surface **30**. The lid **24** is removably couplable to the cooking body **22**. The lid **24** provides selective access to the cooking chamber **26**. When coupled to the cooking body **22**, at least a portion of the inner surface **28** extends across a respective portion of the cooking chamber **26**. The cooking vessel **20** is any suitable cooking vessel including a cooking chamber and lid **24**. As a non-limiting example, the cooking vessel **20** is, but is not limited to, a pot, a pan, a Dutch oven, a slow cooker, a pressure cooker, a skillet, or the like.

[0027] While described in terms of the cooking vessel **20** and the cooking appliance **12** being separate, it will be appreciated that the cooking appliance **12** can be integrally formed with the cooking vessel **20**. As a non-limiting example, the cooking appliance **12** can be a slow cooker including an integrally formed pot defining the cooking chamber **26**. The lid **24** can be a lid to the slow cooker.

[0028] FIG. 2 is a schematic side cross-sectional view of the cooking system **10** of FIG. 1 with the cooking vessel **20** provided along the heating zone **16**. The cooking appliance **12** includes the heating element **50**. The heating element **50** can include a set of coils **70**. The set of coils **70** produce a heat that is transferrable to the cooking vessel **20** within the heating zone **16** when an electric current is supplied the set of coils **70**. The heating element **50** can include any suitable component or device configured to produce a heat that is transferrable to the cooking vessel **20** within the heating zone **16**.

[0029] The lid **24** includes a lid body **32** that extends between the inner surface **28** and the outer surface **30**. The lid body **32** includes a centerline axis **76** extending between the inner surface **28** and the outer surface **30**.

[0030] The cooking body 22 terminates at a top end 23. The cooking body 22 extends a first radial width (W1) between radially opposing sides of the top end 23, with respect to the centerline axis 76, when the lid 24 is coupled to the cooking body 22. The lid 24 is adapted to fit two or more cooking bodies 22 having varying first radial widths (W1). As a non-limiting example, the lid 24 can include a series of steps 46 that are sized to fit the two or more cooking bodies 22. The series of steps 46 are provided along the inner surface 28.

[0031] The lid body 32 includes a handle 34. As used herein, the handle 34 is any suitable feature, component, formation, or device that a user of the cooking vessel 20 can utilize to remove the lid 24 from the cooking body 22. The handle 34 can be provided along any suitable portion of the lid body 32 that is accessible to a user when the lid 24 is coupled to the cooking body 22. As a non-limiting example, the handle 34 is provided along the upper surface 30.

[0032] The lid 24 can include a liquid reservoir 36 provided within the lid body 32. The liquid reservoir 36 is any reservoir, cavity, or chamber within the lid body 32 that is configured to hold a volume of liquid. The liquid can be, but is not limited to, water, wine, broth, stock, or the like. An inlet (not illustrated) can be formed along the lid body 32 to provide access to the liquid reservoir 36. Put another way, the lid body 32 can include an inlet to the liquid reservoir 36 that allows for liquid to be added to the liquid reservoir 36. As a non-limiting example, the handle 34 can be removably coupled to the lid body 32 and act as a cork or cover for the inlet of the liquid reservoir 36. As such, the handle 34 can be selectively removable from the lid body 32 to provide for access to the liquid reservoir.

[0033] The liquid reservoir 36 can have any suitable shape or volume. As a non-limiting example, the volume of the liquid reservoir 36 is selected based on a desired volume of liquid that the lid 24 can hold.

[0034] The lid 24 can include a distribution cavity 40 provided within the lid body 32. The lid body 32 can define a separating wall 98 that separates the liquid reservoir 36 and the distribution cavity 40. The distribution cavity 40 is fluidly coupled to the cooking chamber 26 through a set of outlets 44. The set of outlets 44 are formed along the inner surface 28. The distribution cavity 40 is configured to distribute a liquid from the liquid reservoir 36 to the set of outlets 44. Each outlet of the set of outlets 44 includes an outlet centerline axis 45 extending from the distribution cavity 40 and to the inner surface 28. The outlet centerline axis 45 is parallel to or non-parallel to the centerline axis 76. Each outlet of the set of outlets 44 includes the same or differing construction. As a non-limiting example, at least two outlets can include non-parallel outlet centerline axis 45.

[0035] While illustrated as two separate cavities or chambers, it will be appreciated that the liquid reservoir 36 and the distribution cavity 40 can be a single cavity or chamber formed within the lid body 32.

[0036] The lid 24 can include a valve 38 selectively fluidly coupling the liquid reservoir 36 to the distribution cavity 40. The valve 38 is any suitable valve such as, but not limited to, an electrovalve, a pressure valve, a diaphragm valve, or the like.

[0037] The lid 24 includes a sensor 42. The sensor 42 is any suitable sensor configured to measure at least one of a temperature, a humidity, or a combination thereof of the contents of the cooking chamber 26 or of the cooking chamber 26 itself. The sensor 42 can be provided along a portion of the lid 24 confronting the cooking chamber 26. As a non-limiting example, the sensor 42 can be provided along a portion of the inner surface 28 that confronts the cooking chamber 26.

Alternatively, the sensor 42 can be provided along any suitable portion of the lid 24, the cooking body 22, or a combination thereof where the at least one sensor 42 can accurately measure at least one of the temperature, the humidity, or a combination thereof of the contents of the cooking chamber 26 or of the cooking chamber 26 itself. While only a single sensor 42 is illustrated, it will be appreciated that the cooking vessel 20 can include any number of one or more sensors.

[0038] The lid 24 includes a controller module 54 formed along or provided within the lid body 32, or otherwise communicatively coupled to a respective portion of the lid 24. As a non-limiting

example, the controller module **54** can be provided within the handle **34**. The controller module **54** includes a processor **56**, a memory **58**, and a transceiver **60**. The controller module **54** or processor **56** can be operably or communicatively coupled to various portions of the system described herein, such as, but not limited to, the valve **38**, the sensor **42**, or a combination thereof. The sensor **42** and the valve **38** can be communicatively coupled to the controller module **54** through a first communication line **48** and a second communication line **52**, respectively. The first communication line **48** and the second communication line **52** can be at least one of a physical lines formed within the lid body **32**, wireless communications between the sensor **42** and the valve **38** and the controller module **54**, or a combination thereof. The memory **58** can include random access memory (RAM), read-only memory (ROM), flash memory, or one or more different types of portable electronic memory, such as discs, DVDs, CD-ROMs, etc., or any suitable combination of these types of memory. The controller module **54** or processor **56** can further be configured to run any suitable programs. Non-limiting embodiments of the disclosure can be included wherein, for example, the controller module **54** or processor **56** can also be connected with other controllers, processors, or systems of the cooking system **10**, or can be included as part of or a subcomponent of another controller, processor, or system of the cooking system **10**.

[0039] The transceiver **60** is configured to produce and/or receive a signal **62**. The signal **62** includes data representative of the function of the controller module **54**. As a non-limiting example, the signal **62** can include data representative of a measurement from the sensor **42**. As a non-limiting example, the signal **62** can include a command to the controller module **54** to open or close the valve **38**. The signal **62** can be at least one of sent to, receive from, or a combination thereof a set of external components **64**. The set of external components **64** are any suitable device or component that is external the cooking vessel **20**. As a non-limiting example, the set of external components **64** can include at least one of the cooking appliance **12**, an alarm **66**, an electronic device **68**, or a combination thereof. The alarm **66** can be at least one of a visual alarm, acoustic alarm, haptic alarm, or the like. The electronic device **68** can be at least one of a computer, a tablet, a palm pilot, a mobile phone, a landline, or the like.

[0040] While illustrated as being provided along the lid **24**, the controller module **54** can be provided external the cooking vessel **20**. As a non-limiting example, the controller module **54** can be a controller module of the cooking appliance **12**. As such, the valve **38** and the sensor **42** can be wirelessly coupled to the controller module **54**.

[0041] The lid body **32** can have any suitable cross-sectional area when viewed along a plane extending along the centerline axis **76**. The cross-sectional area of the lid body **32** can be, but is not limited to, circular, rectangular, ovular, triangular, or the like.

[0042] During operation, food items are placed within the cooking chamber **26**. The cooking vessel **20** is positioned within, along, or near the heating zone **16**. The heating element **50** is turned on and a heat from the heating element **50** is transferred to the food items within the cooking chamber **26**. The sensor **42** measures at least one of a temperature, a humidity, or a combination thereof of the cooking chamber **26**. The measurement of at least one of a temperature, a humidity, or a combination thereof of the cooking chamber **26** can be used in a variety of ways. The measurement of at least one of a temperature, a humidity, or a combination thereof of the cooking chamber **26**, via the sensor **42**, will hereinafter be referred to as “the sensed parameters of the cooking chamber **26**”.

[0043] The controller module **54** can determine through the sensed parameters of the cooking chamber **26** if a liquid needs to be added to the cooking chamber **26**. As a non-limiting example, if the sensed parameters of the cooking chamber **26** is indicative of a low humidity within the cooking chamber **26**, at least a portion of the liquid from the liquid reservoir **36** can be automatically released, via a command sent to the valve **38** via the controller module **54**, as a first flow of liquid (F1). The first flow of liquid (F1) flows into the distribution cavity **40**, where the first flow of liquid (F1) is distributed to the set of outlets **44** and into the cooking chamber **26** as a

second flow of liquid (F2). The second flow of liquid (F2), which is ultimately fed to the cooking chamber **26**, is used to increase the moistness of the food items within the cooking chamber **26** and the humidity of the cooking chamber **26** itself.

[0044] The sensed parameters of the cooking chamber **26** can be used to determine a doneness of the food items within the cooking chamber **26**, or otherwise be used to determine how far along in the cooking process the food items are. As a non-limiting example, the doneness of the food items can be determined as a function of a time, humidity of the cooking chamber **26**, temperature of the cooking chamber **26**, a type of food that the food items are, a desired doneness of the food items, or a combination thereof.

[0045] Put another way, the sensed parameters of the cooking chamber **26** will be different based on the food item within the cooking chamber **26**. As a non-limiting example, a moister food item such as a soup or a stew will create a higher humidity within the cooking chamber **26**, while a drier food item such as a steak will create a lower humidity within the cooking chamber **26** in comparison with the moister food item. Both of these food items, however, have a humidity and/or temperature threshold that, when measured by the sensor **42**, indicates that the respective food item needs additional moisture or otherwise that the temperature needs to be adjusted. As such, the controller module **54** can include within the memory **58** or a memory accessible to the controller module **54** a library of food items and what the corresponding sensed parameters of the cooking chamber **26** mean for a given food item. It is contemplated that the library of food items can be preset or loaded onto the memory **58** or the memory accessible to the controller module **54**. It is contemplated that the food library can be based on user preference that are set by a preference of a user of the cooking vessel **20**. As a non-limiting example, a user of the cooking vessel **20** may desire a well-done steak as opposed to a rare steak. It is contemplated that a well-done steak will create a higher temperature and a lower humidity when fully cooked, while a rare steak will be the opposite. As such, the controller module **54** can utilize the sensed parameters of the cooking chamber **26**, and the stored library of food items that can be at least partially set by the user to determine the food items doneness based on the desire of the user. As a non-limiting example, the user may desire that an additional ingredient be added to the food item when a certain level of doneness is achieved. The user can upload this information to the food item library such that the controller module **54** can send an indication to the user when the additional ingredient is ready to be added based on the sensed parameters of the cooking chamber **26**.

[0046] As a non-limiting example, the sensed parameters of the cooking chamber **26** can be used to ensure that the cooking body **22** does not dry out and subsequently burn the food item within the cooking chamber **26**. If the sensed parameters of the cooking chamber **26** are indicative of the fact that the cooking body **22** is drying out, the liquid form within the liquid reservoir **36** can be automatically added to the cooking chamber **26**, or an indication can be sent to the user of the cooking vessel **20** that additional moisture is required within the cooking chamber **26** or that the cooking body **22** is going to dry out. Conversely, the sensed parameters of the cooking chamber **26** can be used to determine if the cooking body **22** is too moist (e.g., there is too much liquid within the cooking body **22**). As such, the controller module **54** can send an indication to the user of the cooking vessel **20** that at least some of the liquid within the cooking chamber **26** needs to be drained from the cooking chamber **26**.

[0047] It is contemplated that based on the sensed parameters of the cooking chamber **26** the controller module **54** can automatically determine the doneness of the food items. The controller module **54** can send an update, via the signal **62**, of the doneness of the food item to the set of external components **64**. As a non-limiting example, the controller module **54** can send an indication to the set of external components **64** that the food items within the cooking chamber **26** are adequately cooked. As used herein, the term “adequately cooked” refers to the food items being fully cooked based on a preset threshold of the sensed parameters of the cooking chamber **26** for a given food item. The preset threshold can be preloaded on the memory **58**, a memory accessible to

the controller module **54** (e.g., the cloud), or be set by a user prior to the cooking of the food item. [0048] As a non-limiting example, a user can set the preset threshold on the cooking appliance **12**, the electronic device **68**, or a combination thereof and the controller module **54** can receive a value of the preset threshold via the signal **62**. As a non-limiting example, the controller module **54** can send an indication to the set of external components **64**, via the signal **62**, that the food items within the cooking chamber **26** are at a certain level of doneness or at a certain point in the cooking process of the food item. As a non-limiting example, the controller module **54** can send an indication to the set of external components **64**, via the signal **62**, that the food items are at a point within the cooking process where an additional ingredient or item needs to be added to the cooking chamber **26**.

[0049] As a non-limiting example, the sensed parameters of the cooking chamber **26** can be used to adjust temperatures of the heating element **50** or heating zone **16**. As a non-limiting example, the sensed parameters of the cooking chamber **26** can indicate that the temperature of the food item or the cooking chamber **26** is too high or too low. As such, the controller module **54** can send a command to or otherwise directly control, via the signal **62**, the operation of the cooking appliance **12** and lower or raise, respectively, an amount of heat generated by the heating element **50**. As a non-limiting example, the cooking vessel **20** can automatically turn off, via the signal **62**, the heating element **50** when it is determined that the food item is fully cooked or nearing fully cooked.

[0050] The cooking vessel **20** allows for the automatic control of the cooking process of the food item within the cooking chamber **26**. The automatic control, done via the controller module **54**, ensures that the food item is not too moist or too dry and not overcooked or undercooked by at least one of adding moisture to the cooking chamber **26**, adjusting a heat output of the heating element **50**, or a combination thereof. The controller module **54** can receive the sensed parameters of the cooking chamber **26**. The controller module **54** can utilize the sensed parameters of the cooking chamber **26** to at least one of determine a doneness of the food item, determine if the temperature within the cooking chamber **26** falls outside of an acceptable range (e.g., the cooking chamber **26** is too hot or too cold), or a combination thereof. The controller module **54** can automatically supply liquid from the liquid reservoir **36** to the cooking chamber **26** based on the received sensed parameters of the cooking chamber **26**. The controller module **54** can automatically directly adjust or send a command to the cooking appliance **12** to adjust a heat output of the heating element **50** based on the received sensed parameters of the cooking chamber **26**. The controller module **54** can automatically send a message to or otherwise indicate to a user of the cooking vessel **20** based on the received sensed parameters of the cooking chamber **26**. The message or indication can be indicative of at least one of a doneness level of the food item, a need for an additional ingredient, a presence of a fire, or a combination thereof.

[0051] While the liquid reservoir **36** and the distribution cavity **40** are shown as being separate cavities formed within the lid body **32**, it is contemplated that the liquid reservoir **36** and the distribution cavity **40** can be a single unitary cavity provided within the lid body **32**. When formed as a single unitary body, the valve **38** can be configured to open or close at least a portion of the outlets of the set of outlets such that fluid can be selectively supplied from the unitary cavity and to the cooking chamber **26**.

[0052] FIG. **3** is a schematic view of the inner surface **28** of the lid **24** as seen from sight-line III-III of FIG. **2**. The inner surface **28** can have any suitable cross-sectional area when viewed along a plane perpendicular to the centerline axis **76** and intersecting the inner surface **28**. The cross-sectional area of the inner surface **28** can be, but is not limited to, circular, rectangular, ovular, triangular, or the like. For purposes of illustration, the series of steps **46** (FIG. **2**) are removed from the inner surface **28**.

[0053] The set of outlets **44** can be arranged in circumferential rows about the centerline axis **76**. The set of outlets **44** can be arranged in any number of one or more rows about the centerline axis **76**. As a non-limiting example, the set of outlets **44** can be arranged along a first row **78**, a second

row **80** and a third row **82**. Each row of the set of rows includes at least one outlet of the set of outlets **44** having an outlet centerline axis **45** at the inner surface **28** provided along the respective row of the set of rows. Two or more rows of the set of rows can be concentric or non-concentric with respect to each other.

[0054] Each row of the set of rows can include any number of one or more outlets of the set of outlet **44**. As a non-limiting example, each row of the set of rows includes two or more outlets of the set of outlet **44**. A total number of outlets of the set of outlets **44** provided along a respective row of the set of rows is equal or non-equal between rows.

[0055] Each outlet of the set of outlets **44** includes a hydraulic diameter (Dh) at the inner surface **28**. The hydraulic diameter (Dh) is the same, different, or a combination thereof between each outlet of the set outlets **44**. As a non-limiting example, a portion of the set of outlets **44** provided along the third row **82** can all be larger than a portion of the set of outlets **44** provided on the second row **80**.

[0056] Each outlet of the set of outlets **44** has a respective cross-sectional area when viewed along a plane extending along the inner surface **28**. The cross-sectional area can be any suitable shape such as, but not limited to, circular, rectangular, trapezoidal, triangular, hexagonal, ovular, or the like. The cross-sectional area between outlets of the set of outlets **44** are the same, different, or a combination thereof. The hydraulic diameter (Dh) varies or remains constant along the outlet centerline axis **45** between the inner surface **28** and the distribution cavity **40**.

[0057] The set of rows can be any suitable shape. The set of rows can be provided along any suitable portion of the inner surface **28**. As a non-limiting example, at least one row of the set of rows can include a respective centerpoint that corresponds to or is offset from the centerline axis **76** at the inner surface **28**.

[0058] The set of rows can be radially spaced from one another a fourth radial distance (W4). The fourth radial distance (W4) is equal or non-equal between adjacent rows of the set of rows. The radial distance (W4) between a first row and a second row radially adjacent to the first row and either circumscribing or circumscribed by the first row can be non-equal along the circumferential extent of the first row and the second row.

[0059] It is contemplated that a positioning of, a size of, and a number of outlets of the set of outlets **44** along the set of rows is selected to ensure that the second flow of liquid (F2) is evenly distributed along the inner surface **28**. As a non-limiting example, at least one of the size of the outlets, the number of outlets, or a combination thereof are used to discourage or encourage the second flow of liquid (F2) from flowing through a respective portion of the lid **24**. As a non-limiting example, a few number of outlets of the set of outlets provided on a specific row, or otherwise smaller outlets provided on the specific row are used to discourage the second flow of fluid (F2) from flowing through said outlets.

[0060] As a non-limiting example, the first flow of fluid (F1) of FIG. 2 is centrally (e.g., radially near the centerline axis **76**) input into the distribution cavity **40**. As such, when the first flow of liquid (F1) will come into contact with the set of outlets **44** provided along the first row **78** prior to coming into contact with the set of outlets **44** provided along the third row **82**. By sizing the outlets of the set of outlets **44** smaller along the inner rows, or otherwise providing a fewer number of outlets of the set of outlets **44** along the inner rows, the first flow of fluid (F1) is discouraged from primarily flowing through the set of outlets **44** provided on the inner rows. Put another way, varying the size or number of outlets of the set of outlets **44** between the set of rows ensures that the first flow of liquid (F1) is evenly distributed within the distribution cavity **40**, thus ensuring that the second flow of liquid (F2) is evenly distribute within the cooking chamber **26**.

[0061] FIG. 4 is a schematic view of a lid **124** of an exemplary cooking vessel **120** suitable for use with the cooking vessel **20** of FIG. 1. The cooking vessel **120** is similar to the cooking vessel **20**; therefore, like parts will be identified with like numeral increased to the 100 series, with it being understood that the description of the cooking vessel **20** applies to the cooking vessel **120** unless

noted otherwise.

[0062] The lid **124** of the cooking vessel **120** includes a lid body **132**. The lid body **132** extends between an inner surface **128** and an outer surface (not illustrated, e.g., the outer surface **30** of FIG. 2). The lid body **132** includes a centerline axis **176**. A set of outlets **144** are provided along the inner surface **128**. Each outlet of the set of outlets **144** includes an outlet centerline axis **145**. The lid **124** is removed from a cooking body (e.g., the cooking body **22** of FIG. 1), however, it will be appreciated that the lid **124** can be couplable to any suitable cooking body. The lid **124** includes a sensor **142** provided along the inner surface **128**. While not illustrated, the lid **124** can include any suitable additional element such as, but not limited to, a controller module (e.g., the controller module **54** of FIG. 2), a liquid reservoir (e.g., the liquid reservoir **36** of FIG. 2), a distribution cavity (e.g., the distribution cavity **40** of FIG. 2), a valve (e.g., the valve **38** of FIG. 2), a handle (e.g., the handle **34** of FIG. 2), a series of steps (e.g., the series of steps **46** of FIG. 2), or a combination thereof.

[0063] The lid **124** is similar to the lid **24** (FIG. 2) in that the lid **124** includes the set of outlets **144** provided along a set of rows. The set of rows of the lid **124**, however, include a greater total number of rows than the set of rows of the lid **24**. As a non-limiting example, the set of rows can include a first row **178**, a second row **180**, a third row **182**, a fourth row **188** and a fifth row **190**. The set of outlets **144** can differ between two or more rows of the set of rows.

[0064] As a non-limiting example the outlets of the set of outlets **144** provided on the third row **182** can be larger than the outlets of the set of outlets **144** provided along the first row **178**. As a non-limiting example, the cross-sectional area of the outlets of the set of outlets **144** provided on the fifth row **190** can be larger than the cross-sectional area of the outlets of the set of outlets **144** provided on the first row **178**, the second row **180**, the third row **182** and the fourth row **188**. As a non-limiting example, at least one outlet of the set of outlets **144** provided on the fourth row **188** can be formed as a non-circular channel **192**, while at least one outlet of the set of outlets **144** provided on the first row **178**, the second row **180**, the third row **182** and the fourth row **188** can be a circular channel.

[0065] As a non-limiting example, at least one row of the set of rows can include outlets of the set of outlets **144** that are circumferentially offset from a radially nearest outlet of the set of outlets **144** provided on a radially adjacent row of the set of rows. As a non-limiting example, at least a portion of the outlets of the set of outlets **144** provided on the fourth row **188** can be circumferentially offset from at least a portion of the outlet of the set of outlets **144** provided on the third row **182**.

[0066] FIG. 5 is a schematic view of a lid **224** of an exemplary cooking vessel **220** suitable for use with the cooking vessel **20** of FIG. 2. The cooking vessel **220** is similar to the cooking vessel **20**, **120** (FIG. 4); therefore, like parts will be identified with like numeral increased to the 200 series, with it being understood that the description of the cooking vessel **20**, **120** applies to the cooking vessel **220** unless noted otherwise.

[0067] The lid **224** of the cooking vessel **220** includes a lid body **232**. The lid body **232** extends between an inner surface **228** and an outer surface **230**. The lid body **232** includes a centerline axis **276**. A liquid reservoir **236** and a distribution cavity **240** can each be provided within the lid body **232**. A separating wall **298** separates the liquid reservoir **236** and the distribution cavity **240**. The lid **224** includes a handle **234**. The lid **224** includes a valve **238** and a sensor **242** communicatively coupled to a controller module **254**. The controller module **254** includes a processor **256**, a memory **258** and a transceiver **260**. The transceiver **260** is configured to produce and/or receive a signal **262**. A set of outlets **244** are provided along the inner surface **228**. The lid **224** is removed from a cooking body (e.g., the cooking body **22** of FIG. 2), however, it will be appreciated that the lid **224** can be couplable to any suitable cooking body.

[0068] The lid **224** is similar to the lid **24** (FIG. 1), **124** (FIG. 4) in that the lid **224** can be configured to distribute a first flow of liquid (F1) to the set of outlets **244**. The lid **224**, however, includes additional structure within the distribution cavity **240** to evenly distribute the first flow of

liquid (F1) to the set of outlets 244. As a non-limiting example, the lid 224 can include a distribution duct 294. The distribution duct 294 can be a singular wall, a plurality of ducts or tubes, or a combination thereof formed within the distribution cavity 240 and being configured to distribute the first flow of liquid (F1) to the set of outlets 244. As a non-limiting example, the distribution duct 294 is a singular wall that extends circumferentially about and radially from the centerline axis 276. The distribution duct 294 can include a set of distribution holes 296 or channels configured to direct the first flow of liquid (F1) into the set of outlet 244.

[0069] Benefits of the present disclosure include a cooking vessel with a decreased burden of use when compared to a conventional cooking vessel. For example, the most basic of conventional cooking vessels require constant supervision of the conventional cooking vessel and further require the users own cooking expertise to determine if the cooking chamber of the conventional cooking vessel is too hot, too cold, too dry, or too moist. In some conventional cooking vessels, the lid of the cooking vessel includes a liquid reservoir that feeds liquid to the cooking chamber over time. These conventional cooking vessels, however, do not react the actual parameters (e.g., temperature and/or humidity) of the cooking chamber. Instead, these conventional cooking vessels put liquid into the cooking chamber based on preset schedule or at a constant pace. The cooking vessel, as described herein, utilizes the sensor to determine at least one of a temperature, a humidity, or a combination thereof of the cooking chamber. The sensed parameters of the cooking chamber can then be used to automatically control the cooking process by at least one of supply a liquid to the cooking chamber, send an indication or message to a user of the cooking vessel (e.g., to indicate the doneness of the food and/or the need for an additional ingredient), adjust a heat output of the heating element of the cooking appliance, or a combination thereof. This function of the cooking system, through use of the cooking vessel and the sensed parameters of the cooking chamber, in turn allow for a more autonomous cooking experience (e.g., requiring less user intervention) than the cooking experience of the conventional cooking vessel. This, in turn, greatly reduces the burden of cooking with the cooking vessel in comparison with the conventional cooking vessel.

[0070] To the extent not already described, the different features and structures of the various aspects can be used in combination with each other as desired. That one feature is not illustrated in all of the aspects is not meant to be construed that it cannot be, but is done for brevity of description. Thus, the various features of the different aspects can be mixed and matched as desired to form new aspects, whether or not the new aspects are expressly described.

[0071] This written description uses examples to disclose aspects of the disclosure, including the best mode, and also to enable any person skilled in the art to practice aspects of the disclosure, including making and using any devices or systems and performing any incorporated methods. While aspects of the disclosure have been specifically described in connection with certain specific details thereof, it is to be understood that this is by way of illustration and not of limitation. Reasonable variation and modification are possible within the scope of the forgoing disclosure and drawings without departing from the spirit of the disclosure, which is defined in the appended claims.

[0072] Further aspects of the disclosure are provided by the subject matter of the following clauses:

[0073] A cooking vessel comprising a cooking body defining a cooking chamber, a lid removably couplable to the cooking body, the lid having a lid body extending between an inner surface confronting the cooking chamber, and an outer surface, the lid body having a centerline axis extending between the inner surface and the outer surface, and a liquid reservoir provided within the lid body, and at least one sensor configured to measure at least one of a humidity or a temperature of the cooking chamber wherein a liquid within the liquid reservoir is selectively supplied to the cooking chamber based on at least one of the measured humidity of temperature of the cooking chamber.

[0074] A cooking appliance comprising a cook surface with a heating zone configured to receive a cooking vessel, the cooking vessel comprising a cooking body defining a cooking chamber, a lid

removably couplable to the cooking body, the lid having a lid body extending between an inner surface confronting the cooking chamber, and an outer surface, the lid body having a centerline axis extending between the inner surface and the outer surface, and at least one sensor configured to measure at least one of a humidity or a temperature of the cooking chamber, wherein a temperature of the heating zone is configured to be adjusted based on at least one of the humidity, the temperature, or a combination thereof of the cooking chamber.

[0075] The cooking vessel of any of the preceding clauses, wherein the lid comprises an actuator configured to selectively supply a liquid within the liquid reservoir to the cooking chamber.

[0076] The cooking vessel of any of the preceding clauses, wherein the actuator is an electrovalve.

[0077] The cooking vessel of any of the preceding clauses, wherein the lid comprises a plurality of outlets provided along the inner surface and being fluidly coupled to the liquid reservoir.

[0078] The cooking vessel of any of the preceding clauses, wherein the lid comprises a distribution cavity formed within the lid body and fluidly coupling the liquid reservoir to the plurality of outlets.

[0079] The cooking vessel of any of the preceding clauses, wherein the lid comprises a distribution duct provided within the distribution cavity.

[0080] The cooking vessel of any of the preceding clauses, wherein the distribution duct is formed as a singular wall extending circumferentially about the centerline axis.

[0081] The cooking vessel of any of the preceding clauses, wherein the plurality of outlets are arranged in a plurality of concentric rows radially spaced with respect to one another, with respect to the centerline axis.

[0082] The cooking vessel of any of the preceding clauses, wherein a first outlet of the plurality of outlets includes a first cross-sectional area when viewed along a plane perpendicular to the centerline axis and intersecting the first outlet, and a second outlet of the plurality of outlets includes a second cross-sectional area, larger than the first cross-sectional area, when viewed along a plane perpendicular to the centerline axis and intersecting the second outlet.

[0083] The cooking vessel of any of the preceding clauses, wherein the first outlet is provided radially inward from the second outlet.

[0084] The cooking vessel of any of the preceding clauses, further comprising a controller module communicatively coupled to the at least one sensor, the controller module configured to receive a signal from the at least one sensor indicative of at least one of the temperature or the humidity of the cooking chamber, and automatically selectively fluidly couple the liquid reservoir to the cooking chamber based on the received signal.

[0085] The cooking vessel of any of the preceding clauses, wherein the controller module is configured to communicate at least one of a doneness or a temperature of the cooking chamber to a user of the cooking vessel through a wireless communication.

[0086] The cooking vessel of any of the preceding clauses, wherein the controller module is configured to alert a user of the cooking vessel when the measured temperature exceeds a user-defined temperature threshold.

[0087] The cooking vessel of any of the preceding clauses, wherein the controller module is provided within the lid body.

[0088] The cooking vessel of any of the preceding clauses, wherein the at least one sensor is provided along the lid body.

[0089] The cooking vessel of any of the preceding clauses, wherein the lid comprises a power source operably coupled to the at least one sensor.

[0090] The cooking vessel of any of the preceding clauses, wherein the at least one sensor comprises both a humidity sensor and a temperature sensor.

[0091] The cooking appliance of any preceding clause, wherein the at least one sensor is provided along the lid.

[0092] The cooking appliance of any preceding clause, wherein the cooking vessel includes a

controller module communicatively coupled to the at least one sensor and the cooking appliance, the controller module being configured to receive a signal from the at least one sensor, the signal being indicative of a temperature of the cooking chamber, and automatically adjust the temperature of the heating zone based on the received signal.

Claims

1. A cooking vessel comprising: a cooking body defining a cooking chamber; a lid removably couplable to the cooking body, the lid having: a lid body extending between an inner surface confronting the cooking chamber, and an outer surface, the lid body having a centerline axis extending between the inner surface and the outer surface; and a liquid reservoir provided within the lid body; and at least one sensor configured to measure at least one of a humidity or a temperature of the cooking chamber; wherein a liquid within the liquid reservoir is selectively supplied to the cooking chamber based on at least one of the measured humidity of temperature of the cooking chamber.
2. The cooking vessel of claim 1, wherein the lid comprises an actuator configured to selectively supply a liquid within the liquid reservoir to the cooking chamber.
3. The cooking vessel of claim 2, wherein the actuator is an electrovalve.
4. The cooking vessel of claim 1, wherein the lid comprises a plurality of outlets provided along the inner surface and being fluidly coupled to the liquid reservoir.
5. The cooking vessel of claim 4, wherein the lid comprises a distribution cavity formed within the lid body and fluidly coupling the liquid reservoir to the plurality of outlets.
6. The cooking vessel of claim 5, wherein the lid comprises a distribution duct provided within the distribution cavity.
7. The cooking vessel of claim 6, wherein the distribution duct is formed as a singular wall extending circumferentially about the centerline axis.
8. The cooking vessel of claim 4, wherein the plurality of outlets are arranged in a plurality of concentric rows radially spaced with respect to one another, with respect to the centerline axis.
9. The cooking vessel of claim 4, wherein: a first outlet of the plurality of outlets includes a first cross-sectional area when viewed along a plane perpendicular to the centerline axis and intersecting the first outlet; and a second outlet of the plurality of outlets includes a second cross-sectional area, larger than the first cross-sectional area, when viewed along a plane perpendicular to the centerline axis and intersecting the second outlet.
10. The cooking vessel of claim 9, wherein the first outlet is provided radially inward from the second outlet.
11. The cooking vessel of claim 1, further comprising a controller module communicatively coupled to the at least one sensor, the controller module configured to: receive a signal from the at least one sensor indicative of at least one of the temperature or the humidity of the cooking chamber; and automatically selectively fluidly couple the liquid reservoir to the cooking chamber based on the received signal.
12. The cooking vessel of claim 11, wherein the controller module is configured to communicate at least one of a doneness or a temperature of the cooking chamber to a user of the cooking vessel through a wireless communication.
13. The cooking vessel of claim 11, wherein the controller module is configured to alert a user of the cooking vessel when the measured temperature exceeds a user-defined temperature threshold.
14. The cooking vessel of claim 11, wherein the controller module is provided within the lid body.
15. The cooking vessel of claim 1, wherein the at least one sensor is provided along the lid body.
16. The cooking vessel of claim 1, wherein the lid comprises a power source operably coupled to the at least one sensor.
17. The cooking vessel of claim 1, wherein the at least one sensor comprises both a humidity

sensor and a temperature sensor.

18. A cooking appliance comprising: a cook surface with a heating zone configured to receive a cooking vessel, the cooking vessel comprising: a cooking body defining a cooking chamber; a lid removably couplable to the cooking body, the lid having a lid body extending between an inner surface confronting the cooking chamber, and an outer surface, the lid body having a centerline axis extending between the inner surface and the outer surface; and at least one sensor configured to measure at least one of a humidity or a temperature of the cooking chamber; wherein a temperature of the heating zone is configured to be adjusted based on at least one of the humidity, the temperature, or a combination thereof of the cooking chamber.

19. The cooking appliance of claim 18, wherein the at least one sensor is provided along the lid.

20. The cooking appliance of claim 18, wherein the cooking vessel includes a controller module communicatively coupled to the at least one sensor and the cooking appliance, the controller module being configured to: receive a signal from the at least one sensor, the signal being indicative of a temperature of the cooking chamber; and automatically adjust the temperature of the heating zone based on the received signal.
